



POLITÉCNICA



Diseño y desarrollo de un entorno para la visualización de información biomecánica de pacientes de fisioterapia

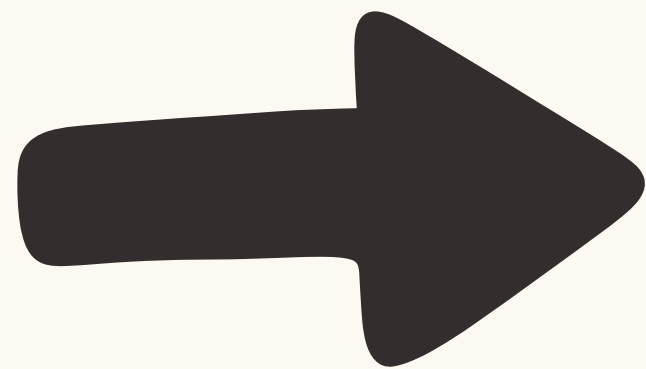
Trabajo Fin de Grado
Grado en Ingeniería Biomédica

Belén Gómez Martínez

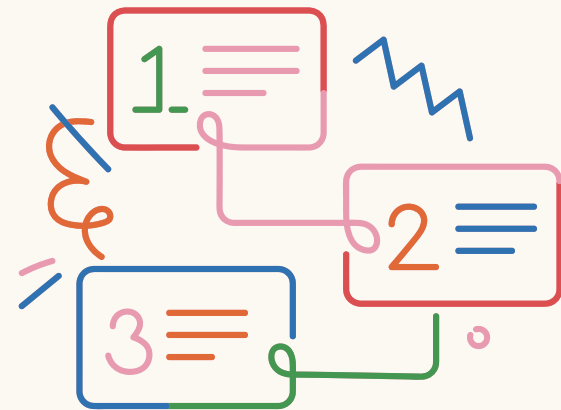
2026







 OpenSim



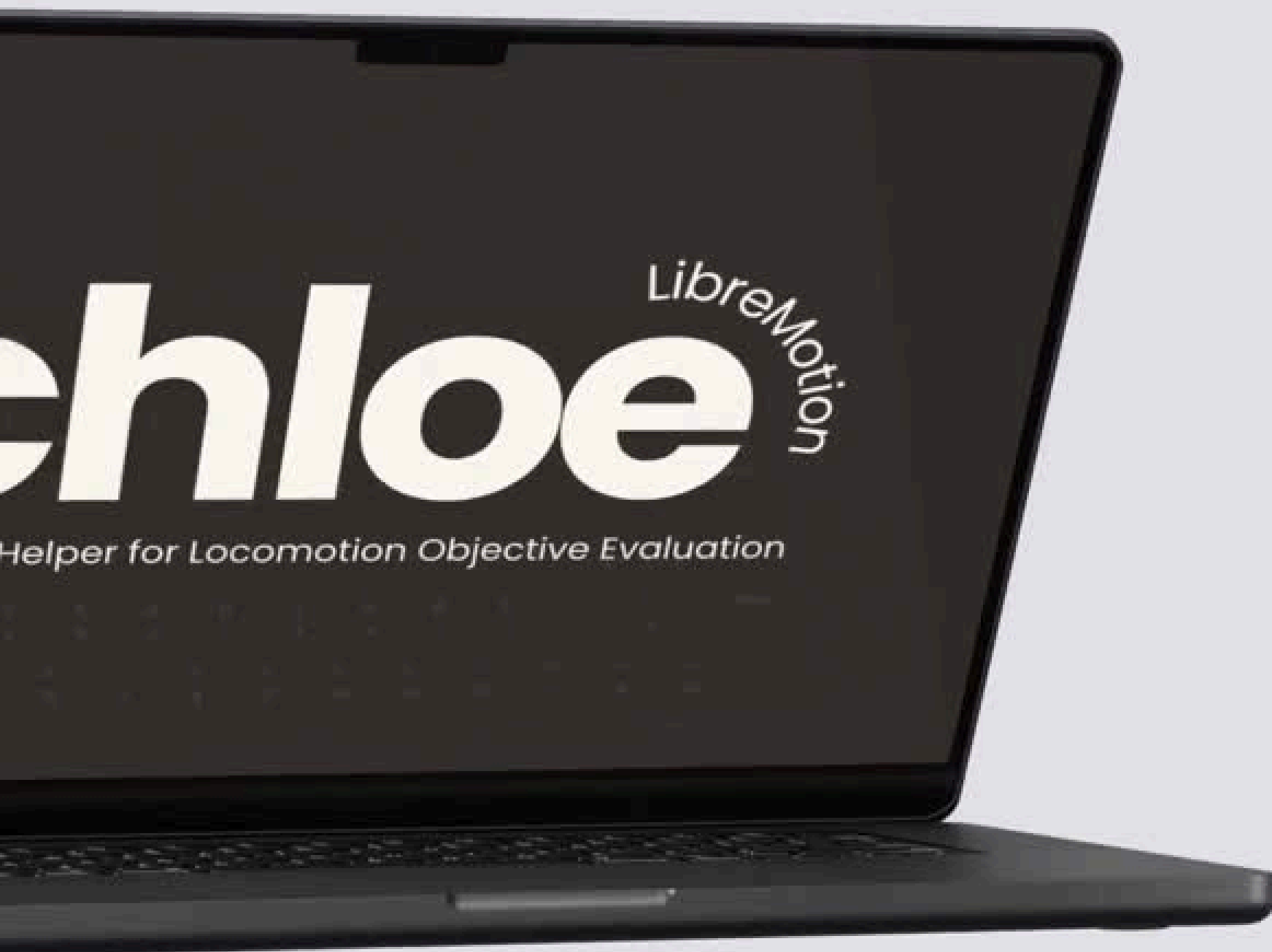
 8000\$
Visual3D

VICON



clinical gap





Requisitos Funcionales (RF)



Requisitos No Funcionales (RNF)

RF1-Ingesta de datos C3D

RF2-Visualización de los marcadores 3D y cambio de visibilidad

RF3-Compatibilidad con múltiples fabricantes (Vicon, OptiTrack, etc.)

RF4-Animación del avatar

RF5-Trazado de señales (EMG/Fuerza)

RF6-Sincronización temporal (animación 3D/datos 2D)

RF7-Multilingüe (ES, EN, FR)

RF8-Extracción de metadatos

RNF1-Portabilidad

RNF2-Buen rendimiento (60 FPS)

RNF3-Robustez ante ausencia de datos

RNF4-Interfaz intuitiva

RNF5-Concurrencia

STACK TECNOLÓGICO DE chloe

LIBREMOTION


python™
ezc3d 
numpy

processor.py

RNF5-Concurrencia

 Gunicorn
4 workers

 Flask
app.py

 docker®

RNF1-Portabilidad

JavaScript



Three.js
Plotly.js

main.js
trajectories.js
kinematics.js
vectors.js
plots.js
i18n.js
constants.js

HTML



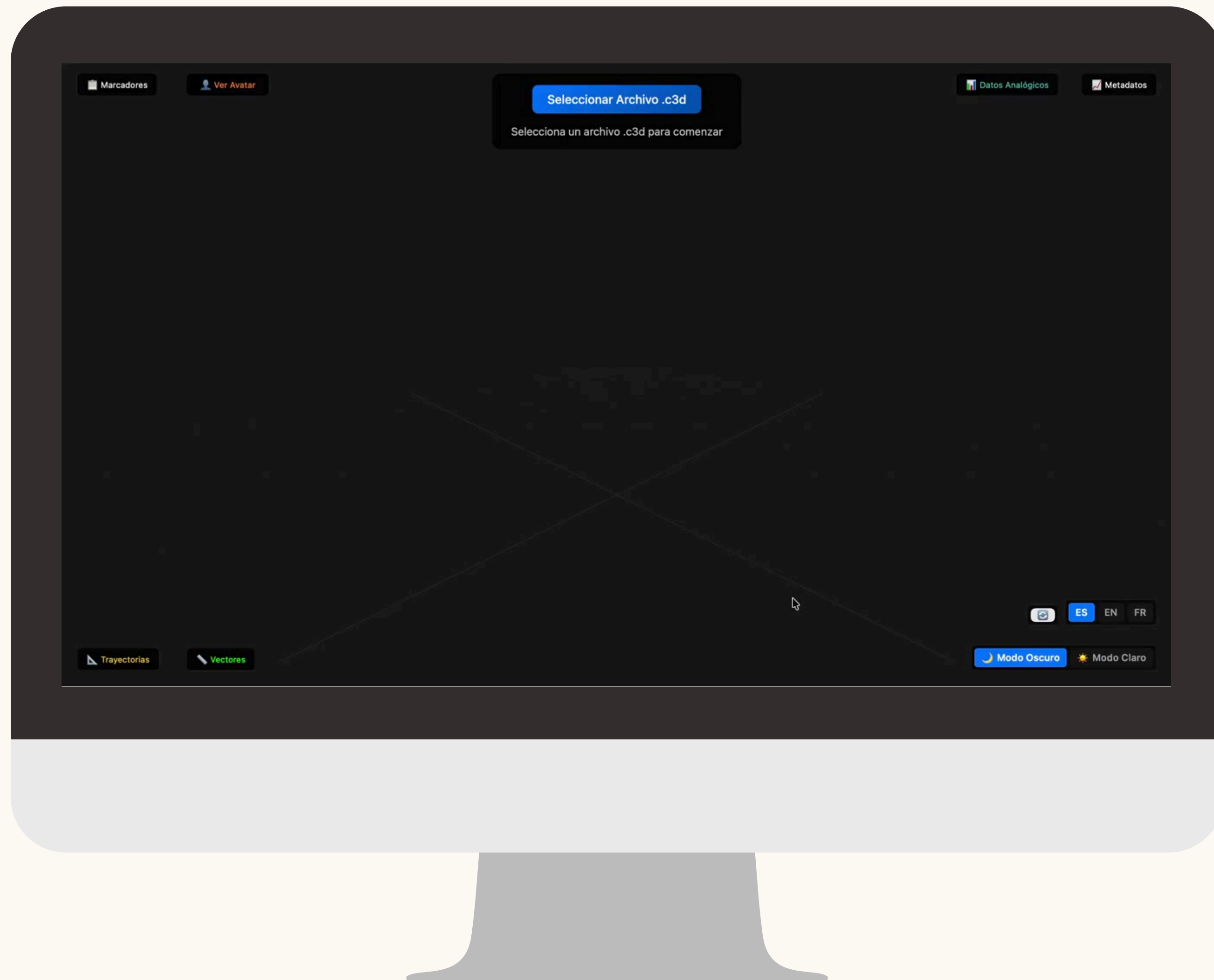
index.html
styles.css

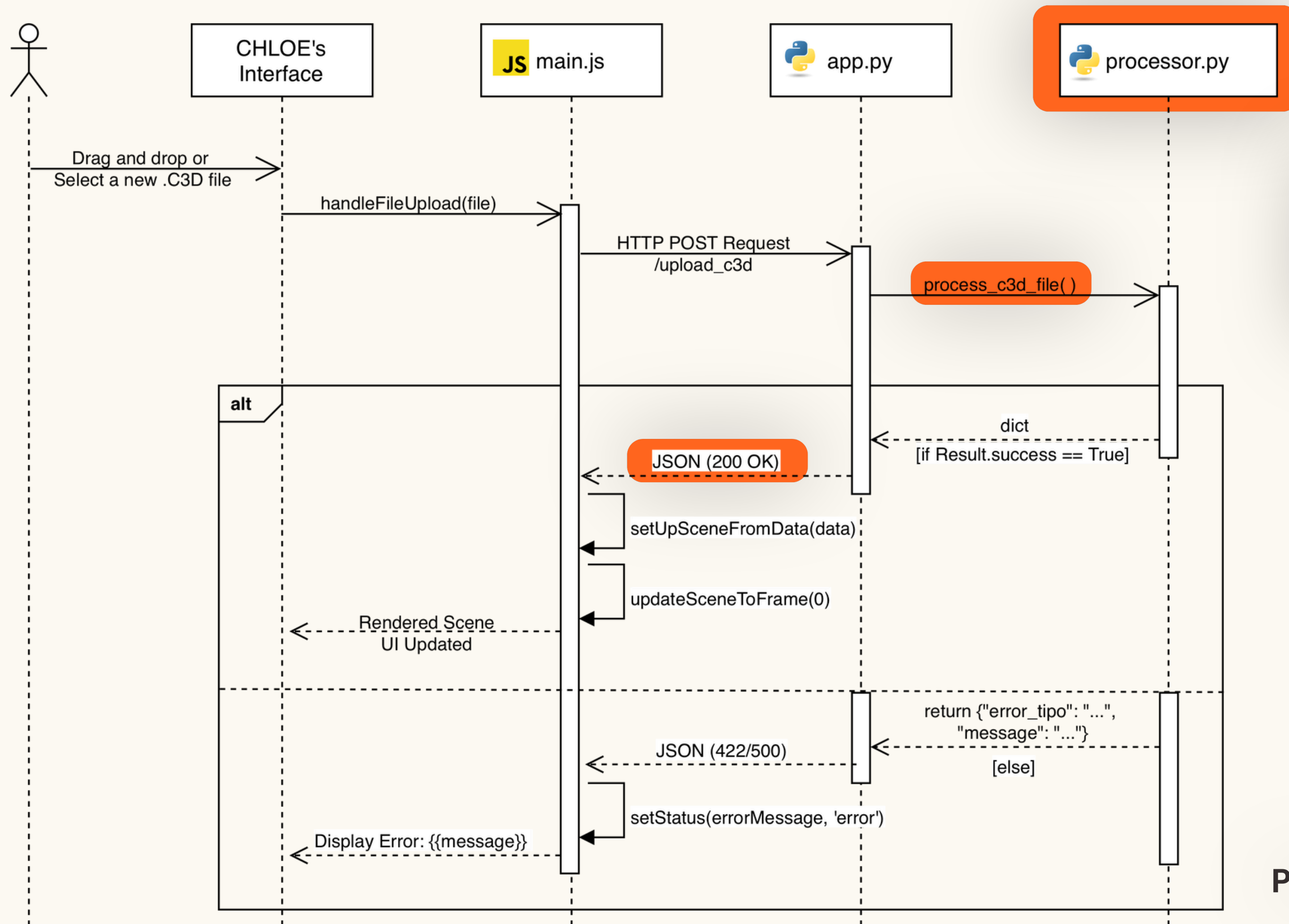
CSS



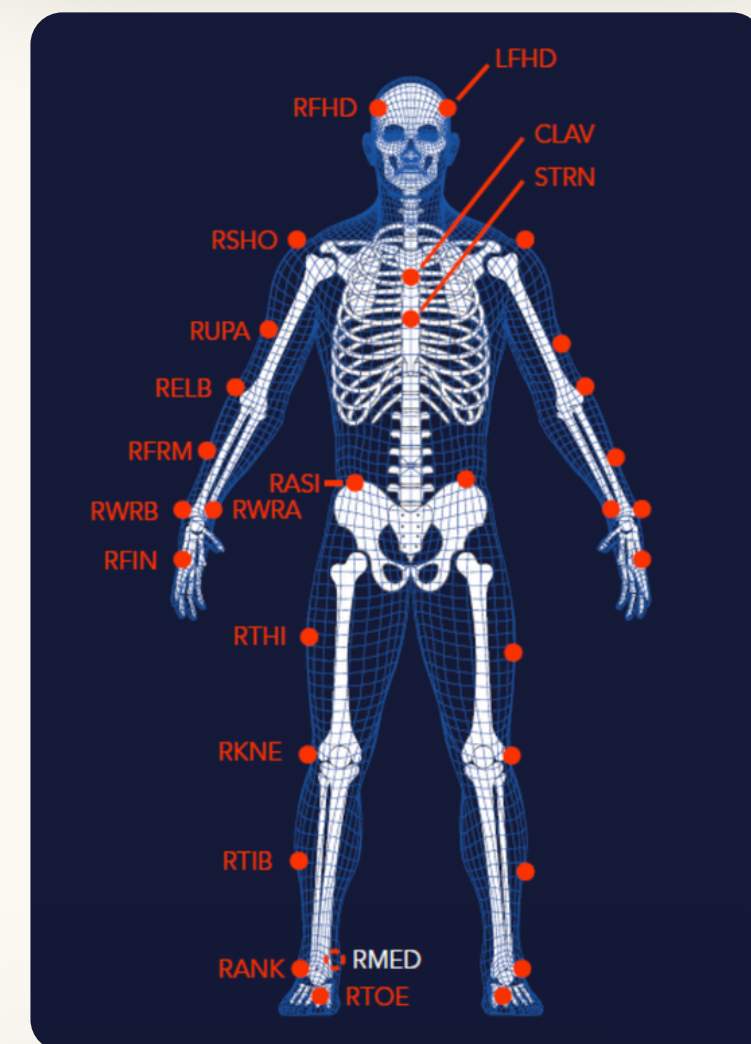
 mixamo

maniqui.fbx

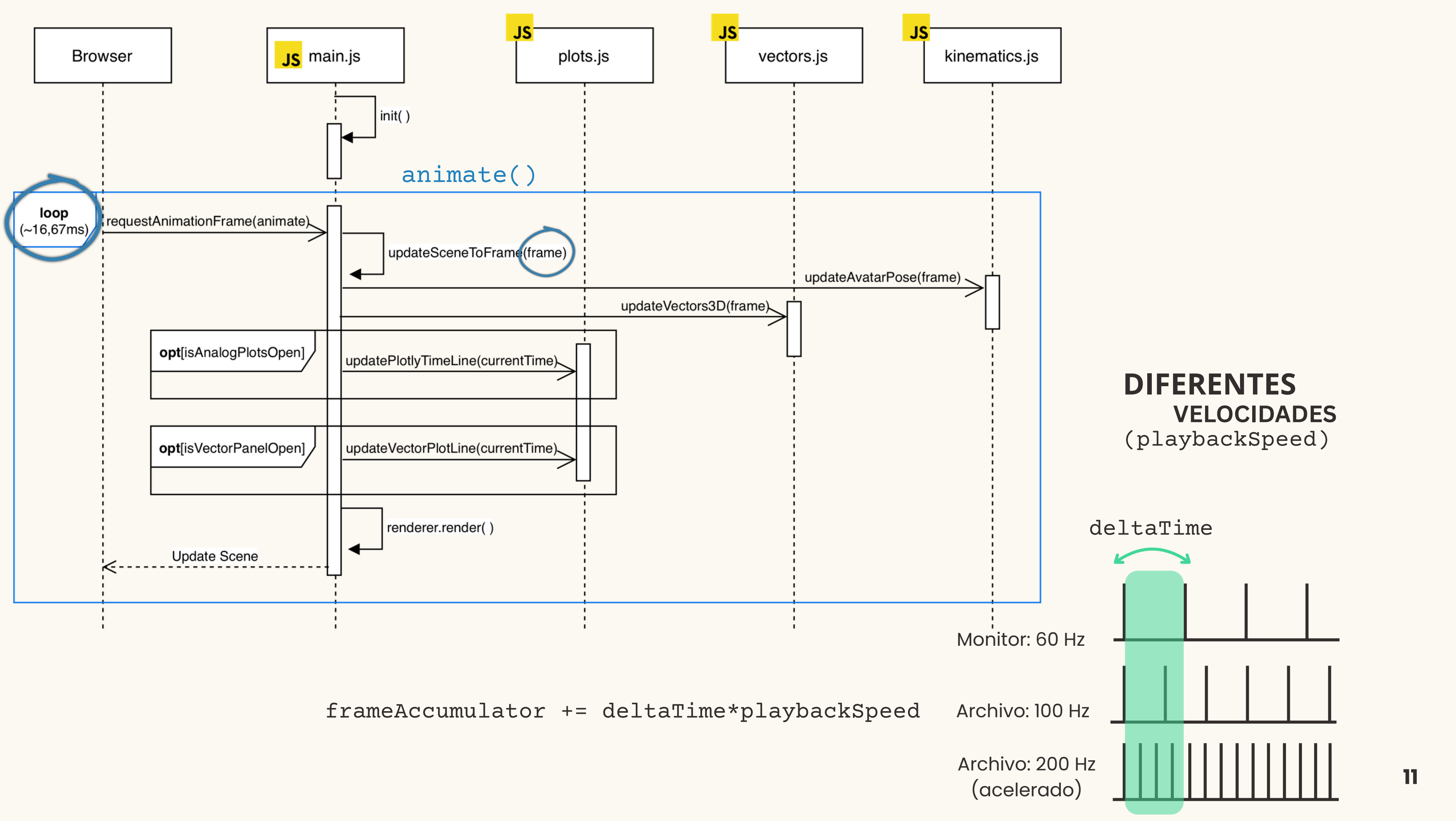


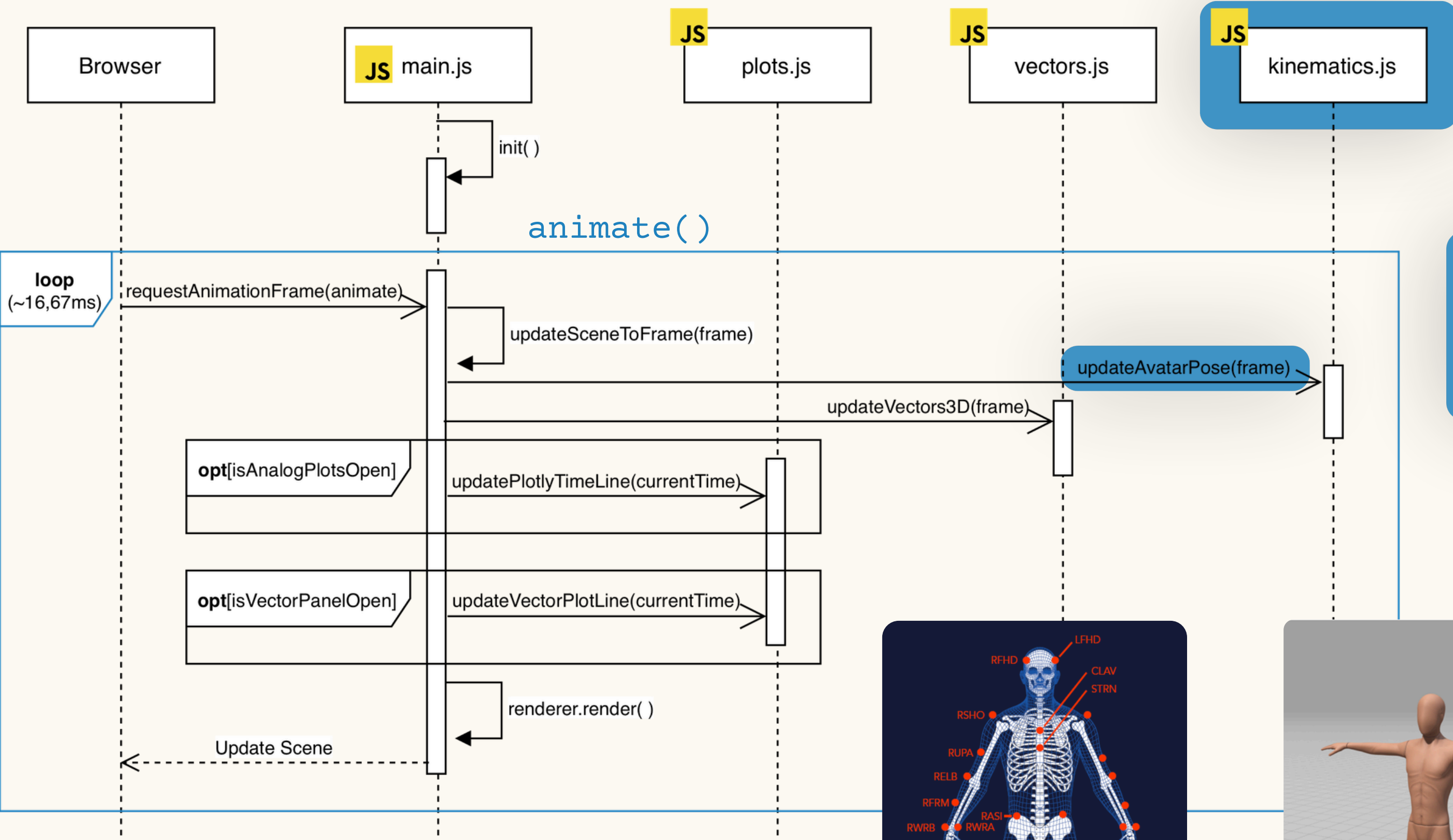


- unidades a metros
- $(X, Y, Z) \rightarrow (X, Z, -Y)$
- STANDARD_MAP

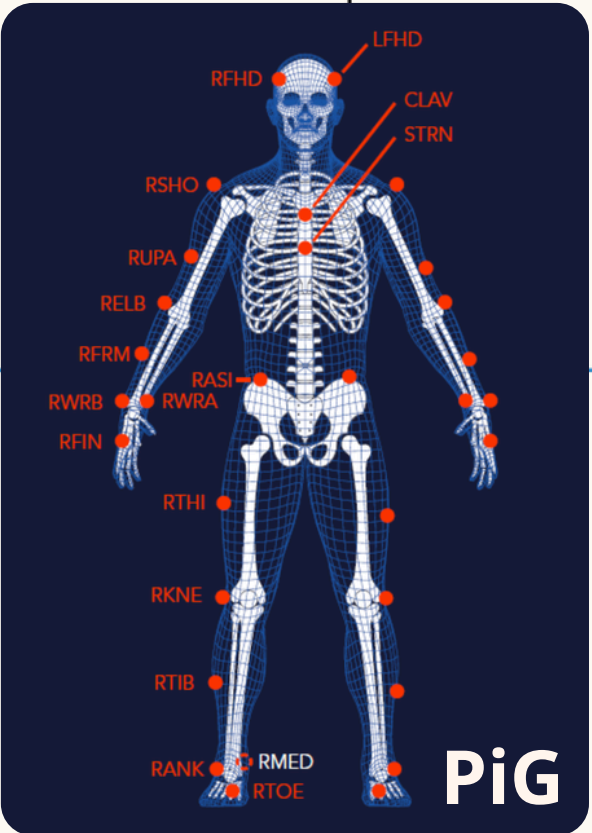


Protocolo Plug-in Gait, Vicon

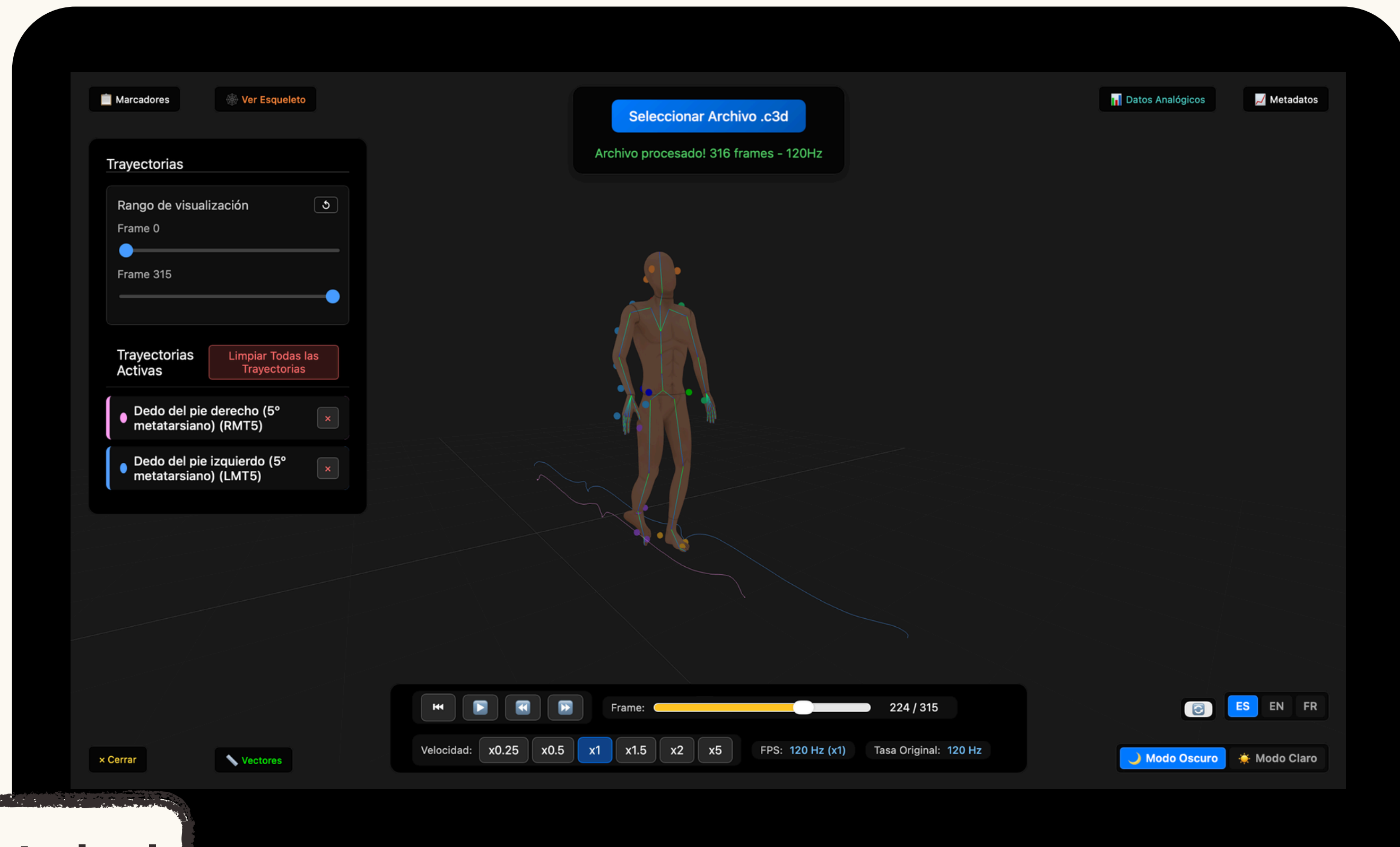




frame = {...,
marker: (x, y, z),
...}



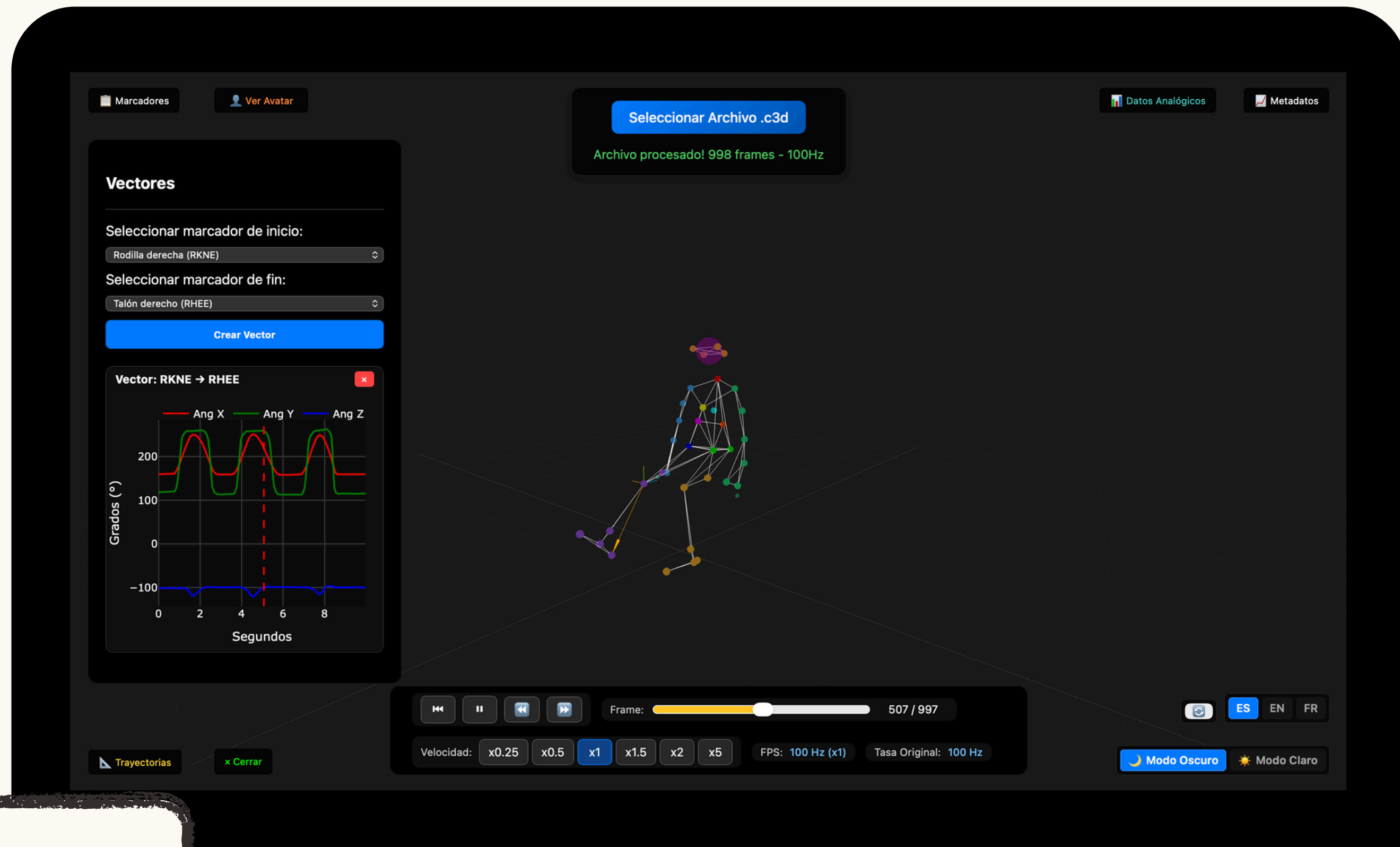
RF4-Avatar animado



¿dónde ha estado el marcador a lo largo del movimiento?

módulo de trayectorias

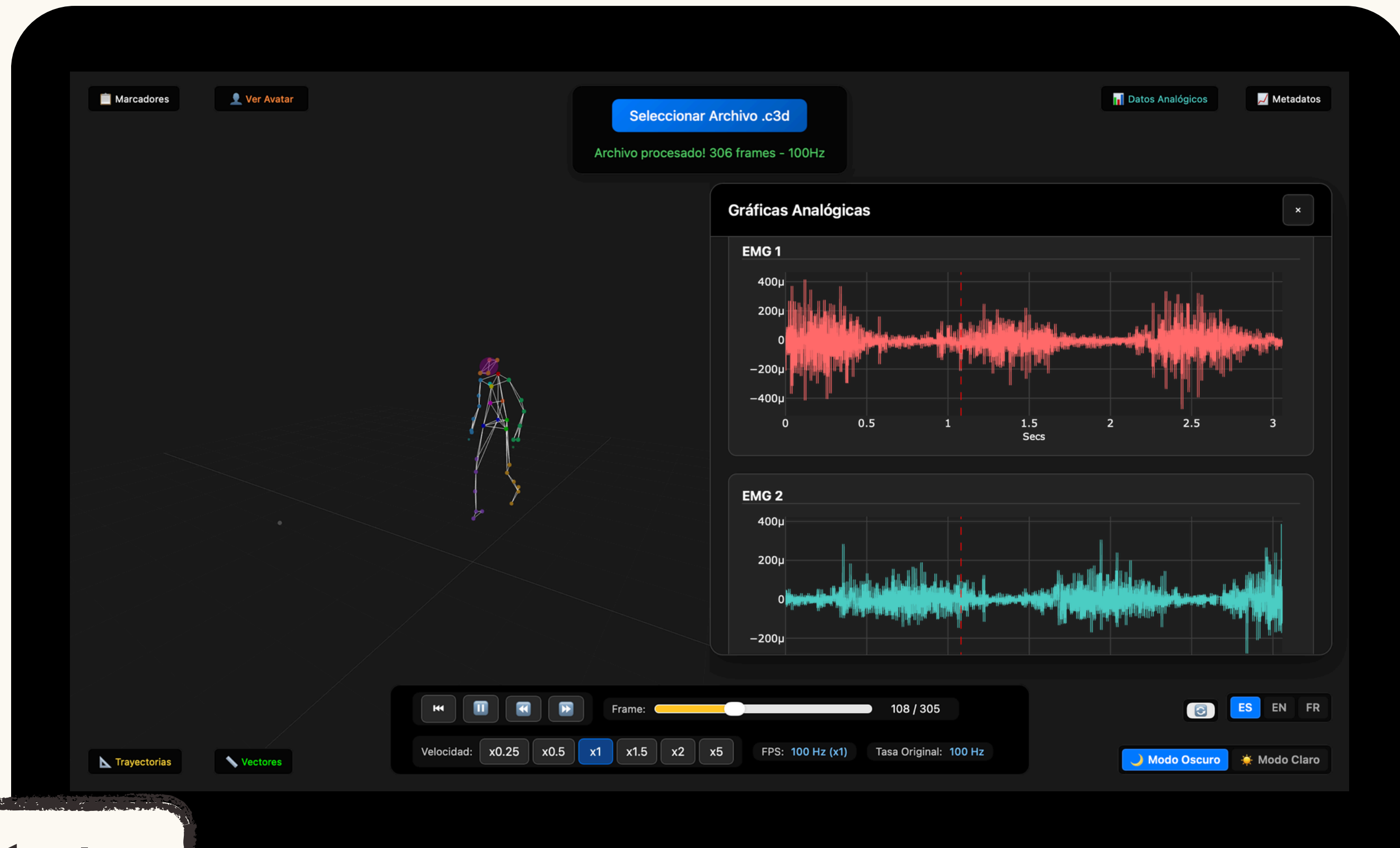
trajectories.js



¿cuánto se inclina
este segmento ?

módulo de vectores

vectors.js & plots.js



¿qué hacía el
músculo en ese
instante ?

módulo de datos analógicos

plots.js

chloe LibreMotion

Clinical Helper for Locomotion Objective Evaluation

VALIDACIÓN DE chloe^{LIBREMOTION}

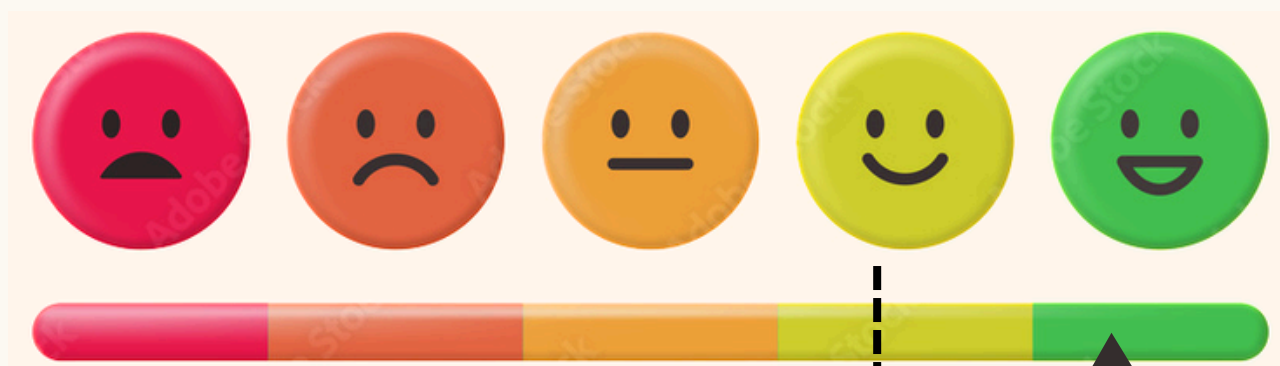


1

VALIDACIÓN
USABILIDAD

2

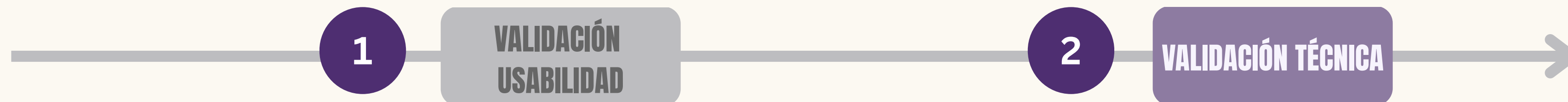
SYSTEM USABILITY SCALE (sus)



85.36

RNF4-Interfaz intuitiva





SUITE OFICIAL DE ARCHIVOS DE PRUEBA



diferentes fabricantes



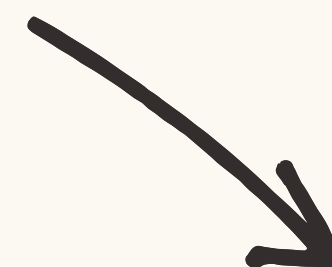
Vicon, MAC, IST, Codamotion, ART,
Cometa, NextGen Ergonomics



diferentes tamaños



100MB, 364 marcadores, 18.178 frames
LABELS2



diferentes arquitecturas



Intel, DEC



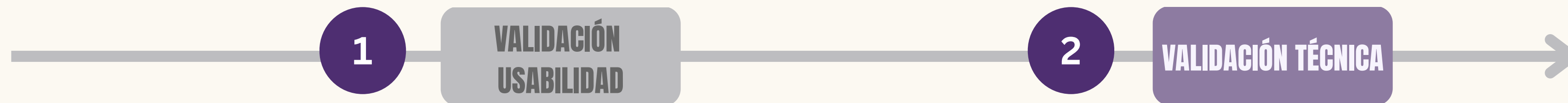
SGI/MIPS



diferentes malformaciones



FRAMES como float
información inválida (ni en POINTS ni en ANALOG)



SUITE OFICIAL DE ARCHIVOS DE PRUEBA



diferentes fabricantes



RF3-Compatibilidad con múltiples fabricantes
RF5-Trazado de señales



diferentes arquitecturas



RF1-Ingesta de C3D



diferentes tamaños



RF2-Visualización de los marcadores
RNF2-Buen rendimiento (60 FPS)



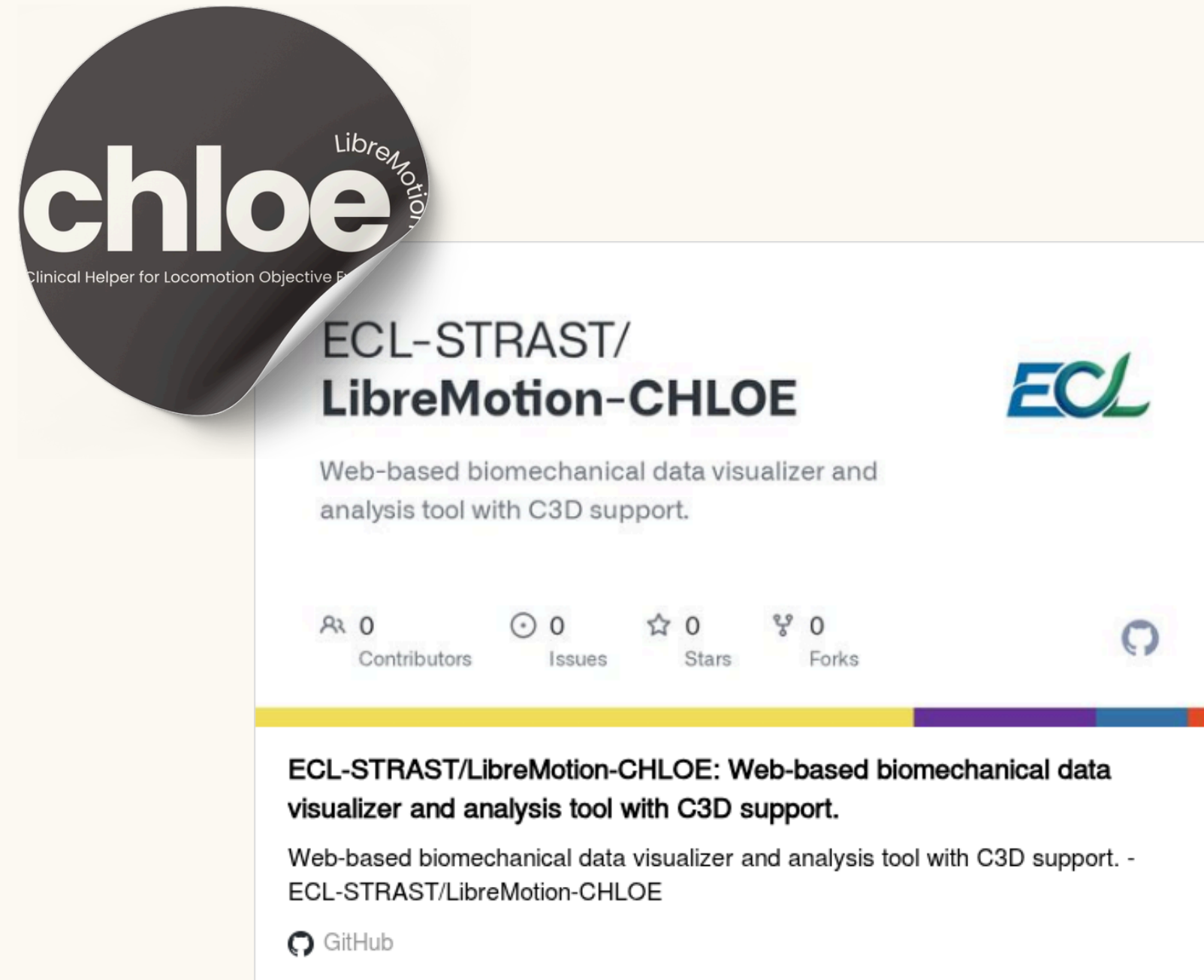
SGI/MIPS



diferentes malformaciones



RNF3-Robustez ante ausencia de datos



CASOS DE USO
NO CONTEMPLADOS AL INICIO...

IMAGINERÍA MOTORA



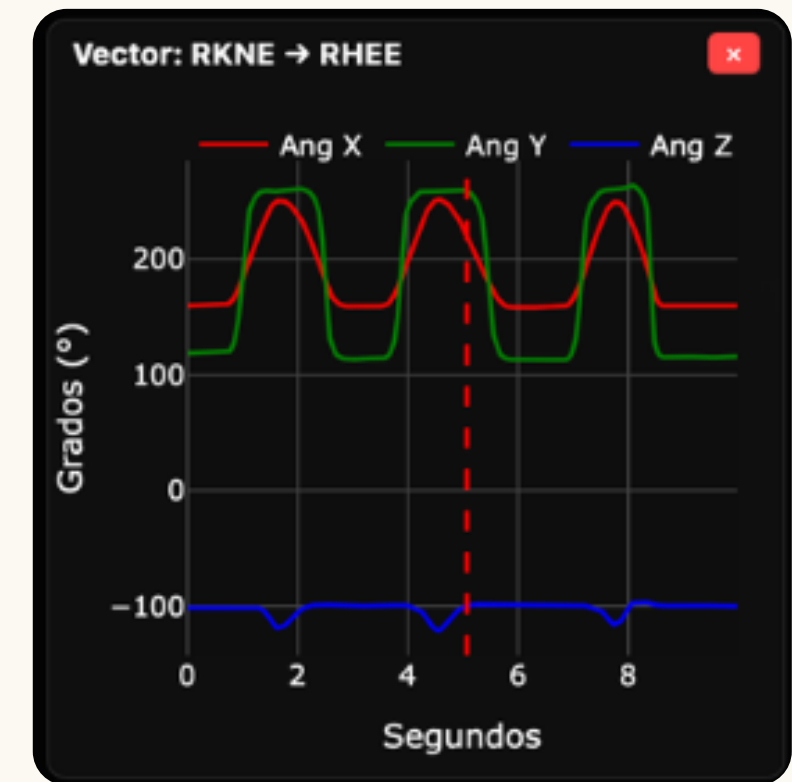
EDUCACIÓN EN BIOMECÁNICA



LÍNEAS FUTURAS



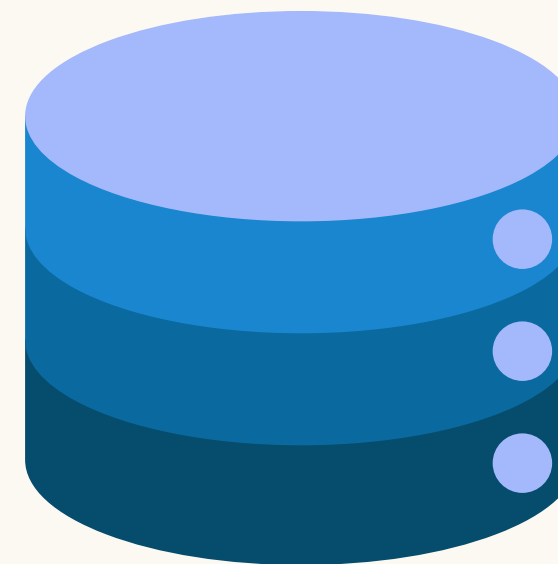
MALLA PERSONALIZADA



VALORES NORMATIVOS



REALIDAD EXTENDIDA



BASE DE DATOS PERSISTENTE

chloe

LibreMotion

Clinical Helper for Locomotion Objective Evaluation

ID	Requirement	Validation Method	Status
<i>Functional Requirements (FR)</i>			
FR1	C3D Ingestion	Test Suite (Sample01, Sample02, Sample08, Sample36)	PASS*
FR2	3D Visualization	Stress Test (Sample12) & Manufacturer Files	PASS
FR3	Multi-Manufacturer	Manufacturer Suite (Sample00: all 7 vendors)	PASS
FR4	Avatar Animation	Clinical Usability Sessions (Skeleton & Avatar Modes)	PASS
FR5	Analog Signal Plotting	EMG/GRF Samples (Vicon, Cometa Workflow)	PASS
FR6	Synchronization	Cyclic Upper-Limb Trial & Clinical Usability Sessions	PASS
FR7	Multi-language	Professional Evaluation (Dynamic UI Switching)	PASS
FR8	Metadata Extraction	Parsing Verification (Metadata Dashboard Display)	PASS
<i>Non-Functional Requirements (NFR)</i>			
NFR1	Portability (Docker)	Multi-Environment Clinician Deployment	PASS
NFR2	Performance	Rendering Benchmarks (60 FPS with Sample12) & Devices	PASS
NFR3	Robustness	Malformed Samples (Sample13, Sample16, Sample18, Sample20, Sample21)	PASS
NFR4	Usability	SUS Questionnaire Evaluation ($N = 7$, Mean: 85.36)	PASS
NFR5	Concurrency	Production deployment with Gunicorn (4 worker processes)	PASS

**Note: FR1 is classified as PASS with the documented exception of SGI/MIPS Big-Endian architectures, as detailed in Section 5.2.*